Power Budget

Overview

This power budget checks that my 9V adapter and 5V regulator can run everything on my Main Control Board at the same time, with a 25% safety margin. It only covers my board, since each teammate has their own adapter and regulator. I used the class power budget template. Everything runs off one 5V rail except the 9V power LED (D6), which sits before the regulator.

A. Major Components

Component	Part number	Supply voltage	Qty	Max current (mA)	Total (mA)
Curiosity Nano (PIC18F57Q43)	DM164150	1.8 to 5.5V	1	170.000	170.000
Dual op amp	MCP6002-I/P	1.8 to 6.0V	1	0.856	0.856
LCD logic	NHD-0216K1Z-FL-YBW	4.5 to 5.5V	1	3.000	3.000
LCD backlight	NHD-0216K1Z-FL-YBW	4.9 to 5.1V	1	156.000	156.000
Cup sensor (LDR + 10k)	PDV-P8103	5V	1	0.516	0.516
Subtotal					**330.372**

B. 5V Rail

Load	Total (mA)
Major components from Section A	330.372
2.5V reference divider (two 10k)	0.258
100k pull-downs driven high: LCD E (R24) plus my 4 ribbon outputs into teammates' 100k pull-downs	0.260
LCD contrast pot (10k)	0.638
3 status LEDs (2.2k each)	7.026
5 button pull-ups (10k each, includes the debug button)	2.576
Debug LED and 5V power LED (2.2k each)	4.684
Total	**345.814**

Load	Total (mA)
Safety margin	25%
Total with margin	**432.268**

C. Regulator

Regulator	Part number	Max current (mA)	Needed (mA)	Remaining (mA)
5V linear regulator	TI LM1086IT-5.0/NOPB	1,500	432.268	**1,067.732**

D. Power Source

Power source	Part number	Output	Max current (mA)	Regulator connected (mA)	Remaining (mA)
Wall adapter	Triad WSU090-2000	9V	2,000	1,500 (LM1086)	**500**

My real draw from the adapter is 444.299mA (432.268mA, plus 10mA for the regulator itself, plus 2.031mA for the 9V power LED), so there's a lot of room left.

E. Battery Life

N/A. My board runs off a wall adapter.

Conclusions

The 2A adapter and the LM1086 can easily power everything on my board with the 25% margin, and nothing comes out negative. The biggest load is the LCD backlight. The regulator turns about 2W into heat, so it gets a heat sink, which keeps it around 71 C (its limit is 125 C).

Microchip only lists the PIC's max current at 3V, not 5V, so I used a safe upper estimate of 170mA for the Nano. The real number should be a lot lower, and I'll measure the actual board current once it's built.

Resources

The full power budget spreadsheet is *here* (also zipped *here*), and a PDF of this page is *here*.

Datasheets used: *LM1086*, *Triad WSU090-2000*, *NHD-0216K1Z-FL-YBW LCD*, *MCP6002*, *PIC18F57Q43*, *PDV-P8103*, *HSE14-384225P heat sink*.